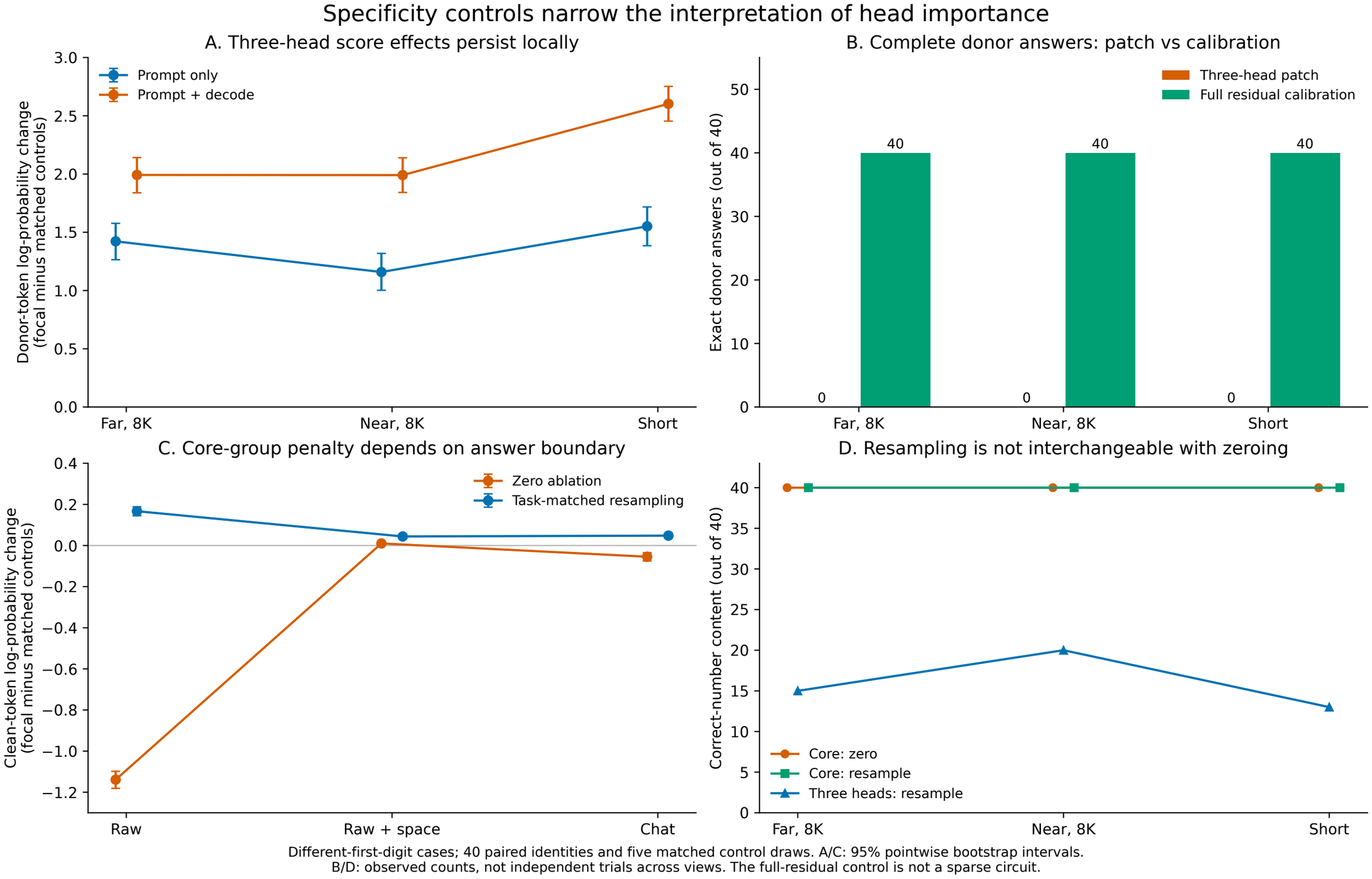
Specificity controls narrow the interpretation of head importance
A. Three-head score effects persist locally
Prompt only
Prompt + decode
Donor-token log-probability change
(focal minus matched controls)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Far, 8K
Near, 8K
Short
B. Complete donor answers: patch vs calibration
Three-head patch
Full residual calibration
Exact donor answers (out of 40)
50
40
30
20
10
0
40
40
40
0
0
0
Far, 8K
Near, 8K
Short
C. Core-group penalty depends on answer boundary
Zero ablation
Task-matched resampling
Clean-token log-probability change
(focal minus matched controls)
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
Raw
Raw + space
Chat
D. Resampling is not interchangeable with zeroing
Correct-number content (out of 40)
40
35
30
25
20
15
10
5
0
Core: zero
Core: resample
Three heads: resample
Far, 8K
Near, 8K
Short
Different-first-digit cases; 40 paired identities and five matched control draws. A/C: 95% pointwise bootstrap intervals.
B/D: observed counts, not independent trials across views. The full-residual control is not a sparse circuit.